جـــامعـــة الـشـــارقـــة
UNIVERSITY OF SHARJAH

College of Computing and Informatics
Department of Computer Science

Course Syllabus

Course Title
Computer Organization and Assembly Language

Course Code
1501252

Spring Semester, 2024/2025

Course Syllabus

Course Title	Computer Organization and Assembly Language
Course Code	1501252
Credit Hours	4
Pre-requisite(s)	Digital Logic Design (0403201)
Co-requisite(s)	None
Semester	Spring — **Year** 2024/2025
Instructors Name	Dr. Ahcene Bounceur
Office Location	W5 – 239
Tel. No.	
Email	abounceur@sharjah.ac.ae
Lecture Times	11:00 am - 12:15 pm Tue, Thu
Office Hours	

Course Description (as in the catalogue):

This course introduces the basic concepts of computer architecture and low-level programming. Subject includes Microprocessors architectures, Bus concepts, 8086 assembly language instruction set, Segmentation and memory addressing modes, debugging and testing programs, Microprocessor systems, pipelining.

Course Objectives/Goals:

The goals of this course are to enable students to:

- Broaden the students' interest and knowledge in topics related to the assembly language. Analyzing, designing algorithms, and implementing programs in assembly language.
- Provide an understanding of programming problems with the machine hardware level. Understanding the control structures, data types including pointers, and concept of modular programming.
- Introduce basic hardware and data representation concepts, including CPU architecture, status flags, and memory mapping.
- Comprehend the programming cycles for high and low-level programming.

Course Learning Outcomes:

By the end of successful completion of this course, the student will be able to:

1. Understand the basic Boolean logic and its applications to programming and computer hardware.
2. Identify the major memory components from the management points of view.
3. Master the translation process from high-level languages to assembly.
4. Understand the implementations of arithmetic expressions, loops, and logic structures at the machine level.
5. Understand error and debugging tracking at the assembly level.
6. Understand the communication methods between application programs and the operating system.
7. Develop skills to write assembly language programs.

Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

Program SLOs	Course SLOs
Apply knowledge of computing and mathematics appropriate to the program's student outcomes and to the discipline.	- Understand the basic Boolean logic and its applications to programming and computer hardware. - Understand the implementations of arithmetic expressions, loops, and logic structures at the machine level.
Analyze a problem and identify and define the computing requirements appropriate to its solution.	- Understand the communication methods between application programs and the operating system.
Design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs.	- Master the translation process from highlevel languages to assembly and to machine level. - Understand the communication methods between application programs and the operating system.
Apply professional, ethical, legal, security and social issues and responsibilities.	- Identify the major memory components from the management points of view. - Understand error and debugging tracking at the machine level.
Communicate effectively with a range of audiences.	Identify the major memory components from the management points of view.
Analyze the local and global impact of computing on individuals, organizations, and society.	- Understand error and debugging tracking at the machine level. - Develop skills to write assembly language programs.

Program SLOs	Course SLOs
Recognize the need for and an ability to engage in continuing professional development.	- Develop skills to write assembly language programs.
Use current techniques, skills, and tools necessary for computing practice.	- Understand the basic Boolean logic and its applications to programming and computer hardware. - Master the translation process from highlevel languages to assembly and to machine level.
Apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices.	- Understand the basic Boolean logic and its applications to programming and computer hardware. - Master the translation process from high-level languages to assembly and to machine level. - Understand the implementations of arithmetic expressions, loops, and logic structures at the machine level.

Weekly Distribution of Course Topics/Contents

Week	Topic	Comments*	CLOs
1. 2.	Computer Systems (Ch 2) Assembly Language: Basic concepts Chapter 2: x86 Processor Architecture	Mile Stone: HW1 Lab# 1	
3. 4.	Memory Components (Cache and internal memory) Chap 4—5 in Stallings Chapter 3: Assembly Language Fundamentals Chapter 4: Data Transfers, Addressing, and Arithmetic	Mile Stone: - Lab# 3&4 - Practical Quiz 1	-
5.		Mile Stone: - Lab#5 - Mile Stone: HW 2, Quiz 1	
6. 7.	Digital Logic for CPU and Memory Chapter 5: Procedures Chapter 6: Conditional Processing	Mile Stone: - Practical Quiz 2, Quiz 2 - Lab# 6, 7, & 8	
8.		Midterm	
9. 10.	Microarctitcture Level (Ch 4 in Tanenbaum) Chapter 7: Integer Arithmetic Chapter 9: Strings and Arrays	Mile Stone: - HW 3 Lab# 9 & 10	-
11.		- Mile Stone: Practical Quiz 3 - Lab# 11	
12. 13.	Chapter 10: Structures and Macros	- Mile Stone: Quiz 3 - Lab# 12 & 13	

14.	More on the Assembly Language and the Instruction Set (Chs 5 & 7)		
15.	Exam Review		
16.	**Final Exam**		

* In the comments, you can add the relevant chapter or notes, etc.

Information on out-of-class assignments with due dates for submission

Assignment/Activity	Due Date	Comments
HW1	Week 4	
HW2	Week 7	
HW3	Week 12	

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**		Date	Weight (%)
Assignments (10) and Quizzes (5)			15
Labs			10
Midterm			25
Practical Quizzes			10
Final Exam	Theory and Practical	Week 16	40
Total			**100**

** You can modify / add other tools relevant to the course.

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators
Understand the basic Boolean logic and its applications to programming and computer hardware.	- Live Online Lectures - Tutorials - TBL	- Quizzes - Exams - Assignments	Marks 60% acceptable target.
Identify the major memory components from the management points of view.	- Live online Lectures - Tutorials - Problem solving	- Quizzes - Exams - Assignments	Marks Project 60% acceptable target.
Master the translation process from high-level languages to assembly and to machine level.	- Live Online - Lectures - Tutorials - Case studies	- Quizzes - Exams - TBL - Assignments	Marks Project 65% acceptable target.
Understand the implementations of arithmetic expressions, loops, and logic structures at the machine level.	- Live online Lectures - Tutorials - Team discussions	- Quizzes - Exams - Project - Assignments	Marks Project 65% acceptable target.
Understand error and debugging tracking at the machine level.	- Live Online Lectures - Tutorials - Projects - Team discussions	- Quizzes - Exams Project	Marks 60% acceptable target.

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators
Understand the communication methods between application programs and the operating system.	- Live online Lectures - Tutorials - TBL	- Quizzes - Exams	Marks 60% acceptable target.
Develop skills to write assembly language programs.	- Case studies		Project Presentation 60% acceptable target.

Teaching and Learning Resources:

Text Book(s):

- Tanenbaum & Austin. (2013). Structured Computer Organization. 6th Edition. ISBN-10: 0132916525 | ISBN-13: 9780132916523, Pearson.
- Stallings, W. (2016). Computer Organization and Architecture. Global Edition, 10th Edition. ISBN13: 9781292096858, Pearson.
- Detmer, R. C. Introduction to 80x86 Assembly Language and Computer Architecture. ISBN:0763717738

Recommended Readings:

- Kip R. IrvineAssembly Language for x86 Processors, Global Edition, 7th Edition, Pearson.

Additional References:

Irvine, K. R. Assembly Language for x86 Processors. 6th Edition. (free).

Other Resources:

Required Teaching Resources:

- Computers with C++ or Java Compiler.
- Visual Studio 2022 Community Edition

Attendance policy:

Attendance is compulsory. A student missing 10% (6 hours) of the total allocated course hours will receive 1st warning notice and a student missing 15% (9 hours) will receive 2nd warning notice. A student missing 20% (12 hours or more) will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Notes:

Teaching Methodology:

A hybrid of traditional lectures for 20-30 minutes followed by short sessions of questions and answers where you (students) are encouraged to answer each other questions. This enforces your ownership for the material taught in class. To achieve this objective, we will use:

In-class presentations: Students are assigned or select topics to present in the class, normally at the end of the semester. Findings are prepared for in-class presentation and discussion. In-Class brainstorming: Students are divided into small teams to examine and solve a particular exercise or provide feedback on some open problems. Discussion follows up among different teams.

Homework Assignments:

Students may work individually or in teams, depending on the topic and the instructor's discretion, on developing computer programs or/and solving a set of problems (problem-based learning technique).

Projects:

Students work individually or in teams on a research topic or implement, design, and test before class presentation and discussion (problem-based and research-based learning).

Exams & Assignments:

There will be one midterm and one final exam carrying a total of 60%, including the practical lab component. The remaining 40% will be divided between the weekly assembly labs (15%), homework and quizzes (15%) and 10% for the programming class project. Late assignments will not be accepted unless prior permission has been granted. If you feel a grade of an exam or any assignment is incorrect or below what you deserve then please see me within one week from obtaining the grade. Exam dates will be announced later in the class.

There are a number of assignments and quizzes that cover all the topics of the course. Students may work in teams on some assignments, which is normally the programming assignments, but must obtain permission first.